

1 Deriving Continuity Equation

Beginning with a control volume, as shown below, it is possible to derive a continuity equation for mass balance.

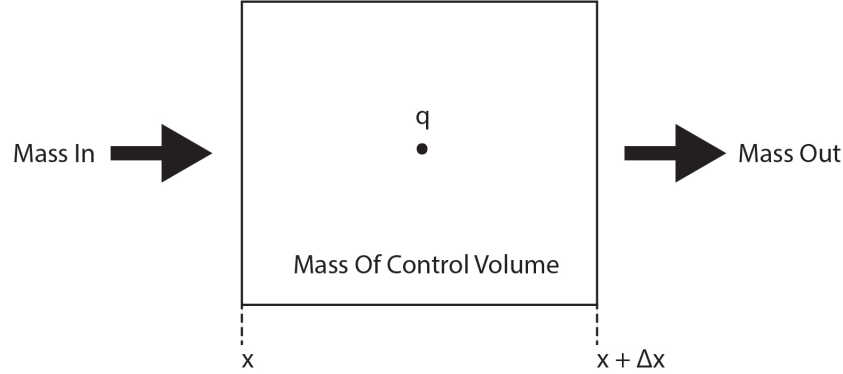


Figure 1: Control volume defined to develop mass balance equation.

Because mass can neither be created or destroyed, the mass conservation for this control volume can be described as

$$\underbrace{\text{Mass Final} - \text{Mass Initial}}_{\text{Accumulation}} = \underbrace{\text{Mass In} - \text{Mass Out}}_{\text{Convection}} + \underbrace{\text{Generation}}_{\text{Source}}, \quad (1)$$

which can be stated as

$$\underbrace{m \cdot V \Big|_{t+\Delta t} - m \cdot V \Big|_t}_{\text{Accumulation}} = \underbrace{F \cdot \Delta t \cdot A \Big|_x - F \cdot \Delta t \cdot A \Big|_{x+\Delta x}}_{\text{Convection}} + \underbrace{q \cdot \Delta t \cdot V}_{\text{Source}}. \quad (2)$$

Here, m is the mass density of the control volume, F is the mass flux, V is the volume of the control volume, Δt is the change in time, and A is the cross-sectional area orthogonal to the direction of Δx . q is in units of $\frac{\text{kg}}{\text{m}^3 \cdot \text{s}}$. By dividing this equation by the Δt and V , one obtains

$$\frac{m}{\Delta t} \Big|_{t+\Delta t} - \frac{m}{\Delta t} \Big|_t = \frac{F}{\Delta x} \Big|_x - \frac{F}{\Delta x} \Big|_{x+\Delta x} + q, \quad (3)$$

which can be reformulated as

$$\frac{m \Big|_{t+\Delta t} - m \Big|_t}{\Delta t} = \frac{F \Big|_x - F \Big|_{x+\Delta x}}{\Delta x} + q. \quad (4)$$

Next, the limit is taken as Δt and Δx go to zero, i.e.,

$$\lim_{\substack{\Delta t \rightarrow 0 \\ \Delta x \rightarrow 0}} \left(\frac{m \Big|_{t+\Delta t} - m \Big|_t}{\Delta t} = \frac{F \Big|_x - F \Big|_{x+\Delta x}}{\Delta x} + q \right), \quad (5)$$

resulting in

$$\frac{\partial m}{\partial t} = -\frac{\partial F}{\partial x} + q. \quad (6)$$

Note that this equation, representing the conservation of total mass, is also called the Continuity Equation. In general form, the equation can be stated as

$$\frac{\partial m}{\partial t} = -\nabla \cdot F + q. \quad (7)$$

The m (mass density of fluid per bulk volume) of the left term can be replaced like

$$\frac{\partial (\phi \rho)}{\partial t} = -\nabla \cdot F + q, \quad (8)$$

where ϕ is porosity and ρ is fluid density. The source term q represents mass injection or production by wells. Positive values correspond to injection wells, while negative values correspond to production wells. The F (mass flux in units of $\frac{kg}{m^3} \frac{m}{sec}$) can be replaced like

$$\frac{\partial (\phi \rho)}{\partial t} = -\nabla \cdot (\rho \cdot u) + q, \quad (9)$$

where u is fluid velocity. Next, by assuming that fluid transport is entirely governed by pressure-driven advection (diffusion is neglected), and that fluid velocity can be described by Darcy's Law, $u = -\frac{k}{\mu} \nabla P$, where k is permeability and μ fluid viscosity,

$$\frac{\partial (\phi \rho)}{\partial t} = -\nabla \cdot \left(\rho \left(-\frac{k}{\mu} \nabla P \right) \right) + q. \quad (10)$$

Considering that viscosity is constant, the continuity equation is,

$$\frac{\partial (\phi \rho)}{\partial t} - \frac{1}{\mu} \nabla \cdot (\rho k \nabla P) = q. \quad (11)$$

2 Finite Volume Method Discretization

Beginning with the continuity equation from mass balance, Equation 11,

$$\frac{\partial (\phi \rho)}{\partial t} - \frac{1}{\mu} \nabla \cdot (\rho k \nabla P) = q, \quad (12)$$

the equation is integrated over a control volume $V_{\langle i,j \rangle}$,

$$\int_{V_{\langle i,j \rangle}} \left[\frac{\partial (\phi \rho)}{\partial t} - \frac{1}{\mu} \nabla \cdot (\rho k \nabla P) \right] dV = \int_{V_{\langle i,j \rangle}} q dV, \quad (13)$$

where i the x -index of the cell, j the y -index of the cell. Applying the divergence theorem converts the volume integral into a surface integral,

$$\frac{\partial (\phi \rho)}{\partial t} V_{\langle i,j \rangle} - \frac{1}{\mu} \int_{A_{\langle i,j \rangle}} \rho k \nabla P \cdot \mathbf{n} dA = Q, \quad (14)$$

where $\mathbf{n}$ is the outward-pointing unit normal vector. Note that $Q = qV_{\langle i,j \rangle}$ is the mass injection rate in units of $\frac{kg}{s}$. The volume of the cell, $V_{\langle i,j \rangle} = \Delta x \Delta y \Delta z$, means this can be discretized as,

$$V_{\langle i,j \rangle} \frac{\phi_{\langle i,j \rangle}^\nu \rho_{\langle i,j \rangle}^\nu - \phi_{\langle i,j \rangle}^n \rho_{\langle i,j \rangle}^n}{\Delta t} - \frac{A_x}{\mu} \left[\rho_{\langle i+\frac{1}{2},j \rangle}^\nu k_{\langle i+\frac{1}{2},j \rangle}^\nu \frac{P_{\langle i+1,j \rangle}^\nu - P_{\langle i,j \rangle}^\nu}{\Delta x} - \rho_{\langle i-\frac{1}{2},j \rangle}^\nu k_{\langle i-\frac{1}{2},j \rangle}^\nu \frac{P_{\langle i,j \rangle}^\nu - P_{\langle i-1,j \rangle}^\nu}{\Delta x} \right] \quad (15)$$

$$- \frac{A_y}{\mu} \left[\rho_{\langle i+\frac{1}{2},j \rangle}^\nu k_{\langle i+\frac{1}{2},j \rangle}^\nu \frac{P_{\langle i,j+1 \rangle}^\nu - P_{\langle i,j \rangle}^\nu}{\Delta y} - \rho_{\langle i,j-\frac{1}{2} \rangle}^\nu k_{\langle i,j-\frac{1}{2} \rangle}^\nu \frac{P_{\langle i,j \rangle}^\nu - P_{\langle i,j-1 \rangle}^\nu}{\Delta y} \right] - Q_{\langle i,j \rangle} = r_{\langle i,j \rangle}^\nu, \quad (16)$$

where n represents the previous time step, ν the current iteration, Δt is the time step, and Δx is the cell length. A_x and A_y are the areas perpendicular to the x and y directions, respectively. Fluid densities in the convection term correspond to cell interfaces, e.g., $i + \frac{1}{2}$, and are upstreamed. Similarly, permeabilities also correspond to cell interfaces, with the harmonic average used, i.e., $k_{i+\frac{1}{2}} = \frac{2k_i k_{i+1}}{k_i + k_{i+1}}$, which guarantees continuity of flux across material interfaces. The residual, r , is defined for each cell. This equation is written for each cell and the system of all of the equations is solved fully implicitly as,

$$r^{n+1} = r^{\nu+1} \approx r^\nu + \frac{\partial r^\nu}{\partial P} \Delta P^{\nu+1} = 0, \quad (17)$$

such that the update to pressure at each iteration, $\Delta P^{\nu+1}$, is found as,

$$\frac{\partial r^\nu}{\partial P} \Delta P^{\nu+1} = -r^\nu. \quad (18)$$

The term $\frac{\partial r^\nu}{\partial P}$ represents the Jacobian. Note that it is defined for each iteration (as opposed to each time step) as the discretization is fully implicit. Once the pressure update is found and applied to create new cell pressures, the residual is rebuilt to check for convergence. Once convergence is achieved the simulator passes to the next time step.

Note that, under no-flow (Neumann) boundary conditions, the normal pressure gradient at the domain boundaries is zero, resulting in zero flux through the boundary faces. Therefore, the boundary flux terms do not contribute to the residuals of the edge cells ($r_{\langle 1,j \rangle}$, $r_{\langle i,1 \rangle}$, $r_{\langle N_x,j \rangle}$, and $r_{\langle i,N_y \rangle}$).

To illustrate how one might construct this system of equations, the following example of a 3×3 system is given,

$$\begin{bmatrix} J_{11} & J_{12} & 0 & J_{14} & 0 & 0 & 0 & 0 & 0 \\ J_{21} & J_{22} & J_{23} & 0 & J_{25} & 0 & 0 & 0 & 0 \\ 0 & J_{32} & J_{33} & 0 & 0 & J_{36} & 0 & 0 & 0 \\ J_{41} & 0 & 0 & J_{44} & J_{45} & 0 & J_{47} & 0 & 0 \\ 0 & J_{52} & 0 & J_{54} & J_{55} & J_{56} & 0 & J_{58} & 0 \\ 0 & 0 & J_{63} & 0 & J_{65} & J_{66} & 0 & 0 & J_{69} \\ 0 & 0 & 0 & J_{74} & 0 & 0 & J_{77} & J_{78} & 0 \\ 0 & 0 & 0 & 0 & J_{85} & 0 & J_{87} & J_{88} & J_{89} \\ 0 & 0 & 0 & 0 & 0 & J_{96} & 0 & J_{98} & J_{99} \end{bmatrix}^\nu \begin{bmatrix} \Delta P_1 \\ \Delta P_2 \\ \Delta P_3 \\ \Delta P_4 \\ \Delta P_5 \\ \Delta P_6 \\ \Delta P_7 \\ \Delta P_8 \\ \Delta P_9 \end{bmatrix} = - \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \\ r_6 \\ r_7 \\ r_8 \\ r_9 \end{bmatrix}^\nu. \quad (19)$$

It is assumed that the numbering convention is $k = i + N_x(j - 1)$. J_{14} is equivalent to writing $\frac{\partial r_{\langle 1,1 \rangle}^\nu}{\partial P_{\langle 1,2 \rangle}}$, noting that $k = 4$ corresponds to index $\langle 1,2 \rangle$ for this 3×3 system. In other words, J_{14} is the first cell's residual's ($r_{\langle 1,1 \rangle} = r_1$) dependence on the pressure in the fourth cell, $P_{\langle 1,2 \rangle} = P_4$.

2.1 Wells

Rate-prescribed wells

In the case that the rate of the well is to be prescribed, the term Q_i is simply added to the residual of the cell in which the well is located as show above. This well, or source term, has no explicit contribution to the Jacobian as it is a prescribed mass rate, independent of the pressure in the cell.

Pressure-prescribed wells

In the case that the pressure of the well is to be prescribed, the term Q_i is found as,

$$Q_i = WI \frac{\rho_i k_i}{\mu} (P_{\text{inj}} - P_i), \quad (20)$$

where P_{inj} is the prescribed injection pressure and ρ is calculated based on P_{inj} (meaning its derivative with respect to the cell pressure is zero). WI is the well index and can either be prescribed or found as,

$$WI = \frac{2\pi h}{\ln\left(\frac{r_e}{r_w}\right) + s}, \quad (21)$$

where h is a well height, r_e is an effective wellbore radius, r_w is the actual wellbore radius, and s is a skin factor. In this case, the derivative of the well term becomes,

$$\frac{\partial Q_i}{\partial P_i} = -WI \frac{\rho k}{\mu}. \quad (22)$$

When permeability is pressure dependent, the derivative becomes,

$$\frac{\partial Q_i}{\partial P_i} = WI \frac{\partial k_i}{\partial P_i} \frac{\rho_i}{\mu} (P_{\text{inj}} - P_i) - WI \frac{k_i \rho_i}{\mu}, \quad (23)$$

for injecting wells. For producing wells, this becomes,

$$\frac{\partial Q_i}{\partial P_i} = WI \frac{\partial k_i}{\partial P_i} \frac{\rho_i}{\mu} (P_{\text{inj}} - P_i) - WI \frac{\rho_i k_i}{\mu} + WI \frac{k_i}{\mu} \frac{\partial \rho_i}{\partial P_i} (P_{\text{inj}} - P_i), \quad (24)$$

as the fluid density is upstreamed.

2.2 Fluid density

For a slightly compressible fluid, the fluid compressibility is defined as

$$c_f = \frac{1}{\rho} \frac{\partial \rho}{\partial P}. \quad (25)$$

Assuming that c_f is constant (so that the fluid is slightly compressible),

$$\frac{1}{\rho} d\rho = c_f dP. \quad (26)$$

Integrating between a reference state (P_0, ρ_0) and the current state (P, ρ) gives

$$\int_{\rho_0}^{\rho} \frac{1}{\rho} d\rho = c_f \int_{P_0}^P dP, \quad (27)$$

which yields

$$\ln \left(\frac{\rho}{\rho_0} \right) = c_f (P - P_0). \quad (28)$$

Exponentiating both sides results in

$$\rho = \rho_0 e^{c_f (P - P_0)}, \quad (29)$$

which is used to calculate the fluid densities in each cell.

2.3 Porosity

For a slightly compressible pore space, the pore compressibility is defined as

$$c_\phi = \frac{1}{\phi} \frac{\partial \phi}{\partial P}. \quad (30)$$

Assuming that c_ϕ is constant (so that the pore space is slightly compressible),

$$\frac{1}{\phi} d\phi = c_\phi dP. \quad (31)$$

Integrating between a reference state (P_0, ϕ_0) and the current state (P, ϕ) gives

$$\int_{\phi_0}^{\phi} \frac{1}{\phi} d\phi = c_\phi \int_{P_0}^P dP, \quad (32)$$

which yields

$$\ln \left(\frac{\phi}{\phi_0} \right) = c_\phi (P - P_0). \quad (33)$$

Exponentiating both sides results in

$$\phi = \phi_0 e^{c_\phi (P - P_0)}, \quad (34)$$

which is used to calculate the porosity in each cell.

2.4 Permeability

Different relationships can be used to describe the dependence of permeability on pressure. In this work, a simple exponential model is adopted,

$$k = k_0 e^{c_k(P-P_0)}, \quad (35)$$

where k_0 is the reference permeability at pressure P_0 , and c_k is a permeability-pressure coefficient. This expression allows permeability to increase or decrease with pressure depending on the sign of c_k .

2.5 Fluid velocity

The fluid velocity across the interfaces can then be found using Darcy's Law as,

$$u = -\frac{k}{\mu} \nabla P. \quad (36)$$

Applying a finite-difference approximation at interface $\langle i + \frac{1}{2}, j \rangle$, for example, gives

$$u_{\langle i+\frac{1}{2}, j \rangle} = -\frac{k_{\langle i+\frac{1}{2}, j \rangle}}{\mu} \left(\frac{P_{\langle i+1, j \rangle} - P_{\langle i, j \rangle}}{\Delta x} \right), \quad (37)$$

with the mass flux $F_{\langle i+\frac{1}{2}, j \rangle} = \rho_{\langle i+\frac{1}{2}, j \rangle} u_{\langle i+\frac{1}{2}, j \rangle}$. Note that for the domain boundaries the velocities are,

$$u_L = 0, \quad u_R = 0, \quad u_T = 0, \quad u_B = 0. \quad (38)$$

This is because the boundary condition used here is no-flow (Neumann).